Architecture: a controlled baseline and structural ladders

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §4 separates fixed settings, prescribed scaling rules, and independent experimental variables. An architecture ladder is interpretable only when it states which of these roles each setting plays.

Concrete empirical starting point

Read the width/depth and μP evidence guide for exact published configurations. AI2's ladder uses widths 768/1,024/1,536/2,048 with depths 12/16/16/16; its target depths are 32/40. This is a task-prediction experiment, not an isolated depth-optimization sweep. The guide compares that practice with fixed-depth μP transfer and explicit shape studies. Bhagia et al., v2, §2.1/Table 1.

Step-by-step protocol

1. Start from a target-compatible decoder baseline. Freeze tokenizer, normalization, activation, positional encoding, embedding tying, attention family, context length, precision, and initialization.
2. Declare a width/depth progression. For each configuration, record hidden width, layer count, head dimension, attention groups, MLP dimension, trunk parameters, and output-head cost.
3. Sweep aspect ratio at matched parameter or compute budgets as a separate ablation. Retune enough to avoid mistaking optimization difficulty for inferior architecture.
4. For MoE, separate active-size scaling, expert-count/sparsity scaling, and expert-granularity sweeps. Keep routing, balancing, shared experts, and drop behavior explicit.
5. Measure throughput and memory on the intended hardware. Compare equivalent quality at a training and serving cost the project can afford.
6. Validate the winning structure at a larger scale and at the intended token horizon and precision.

For a rough dense baseline, $N_{\text{body}} \approx 12d^2n$. Holding N_{body} fixed means increasing width d forces depth n downward under this approximation. This changes sequential computation and attention's relative cost, so equal parameter counts do not imply equal wall-clock time.

Minimal run matrix

Family	Size tiers	Structural variable	Controls
Dense baseline	Three fitting tiers	Prescribed width/depth rule	Fixed primitives and token horizon
Aspect-ratio ablation	Two or more tiers	Width/depth at matched budget	Comparable tuning
MoE candidate	Separate tiers	One of sparsity or granularity	Fixed active-size convention
Holdout	Larger unseen tier	Selected complete architecture	Frozen recipe

This matrix is an illustrative planning template, not a budget recommendation or a published optimum.

Acceptance checklist: comparable compute accounting; no boundary winner left unexplored; numerical stability; downstream task floors; memory and throughput constraints; fresh holdout after recipe selection.

Sources: blog §4 and §5.6; Gemstones, 2025; Krajewski et al., 2024; OLMoE, 2024.

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